

# CRACKING THE

## CODE OF FINANCIAL QA

AI/ML IS REDEFINING TEST AUTOMATION



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# How AI/ML is Redefining Test Automation in Finance



## AI/ML-Driven Enhancements for Finance QA

Building Trustworthy, Compliant, and Scalable Test Systems

### Why It Matters Now:

- Digital banking growth = more frequent releases
- Cyber fraud and KYC failures are rising
- Traditional QA tools aren't keeping up with hybrid tech stacks



### This Presentation Will Show You:

- What gaps current tools fail to solve
- How AI/ML can fix them (feasibly)
- Real-world architecture + tools you can use



## ! Unique Challenges in Financial QA

### 🧬 Different Systems Together:

Old tech like COBOL + new tech like APIs, React, and mobile apps — all in one system.

### 🔑 Strict Rules to Follow:

Finance apps must follow many rules like PCI-DSS, GDPR, SOC2, AML, and PSD2.

### 🕒 No Room for Downtime:

Banking and trading apps must work 24/7 — no breaks for testing.

### 📊 Need for Clean Logs:

Test logs must be safe, clear, time-stamped, and track who did what.

### 🔒 Very Secure Steps:

Tests must handle things like OTPs, fingerprint login, KYC steps, and payment flows (like Stripe 3D Secure).

🎯 **Traditional QA** tools aren't built for this complexity.

🔄 AI/ML offers explainability, speed, & coverage in regulated workflows.



securing **credit/debit card** transactions



protecting **personal data** and **privacy**



five trust principles  
**security, availability, processing integrity, confidentiality, privacy**



**laundering** and **terrorist financing**

**PSD2**

**open banking** more accessible



**cybersecurity, protected access, or regulatory alignment**



## AI-Powered Log Summarization

→ Quickly find key errors in long test logs

## Flaky Test Detection

→ Flag tests that fail randomly & waste CI time



## Test Coverage Insights

→ Detect missing tests in critical areas



## Test Case Generation from Specs

→ Generate test steps directly from requirements



## Explain Test Failures (XAI)

→ Understand why a test failed, not just that it failed



## Smart Test Prioritization

→ Run most important tests first using risk data

| What AI Does                  | Tools You Can Try      | Why It's Useful in Finance                  |
|-------------------------------|------------------------|---------------------------------------------|
| Find issues in test logs      | LogAI, GPT, LangChain  | Saves time, helps with audit reports        |
| Catch flaky tests             | Launchable, sklearn    | Avoid false failures, reduce re-runs        |
| Spot missing test areas       | SonarQube, Embeddings  | Make sure key features are tested           |
| Write tests from requirements | GPT-4, Claude, Testim  | Quick test writing for new features         |
| Explain why tests failed      | SHAP, LIME, Grafana    | Understand problems better, good for audits |
| Run important tests first     | Launchable, CI Plugins | Catch big bugs early, save test time        |

# What QA Tools Must Improve



| What QA Tools Must Improve       | Why It Matters for Finance QA                                 | Feasibility for Non-Tech Users          | Suggested Tech Stack/Tools                                    | Design Step / Implementation LLD                                  |
|----------------------------------|---------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------|
| Work Without Internet            | Banks often work in secure, air-gapped networks               | ⚠ Needs IT team support                 | Jenkins (self-hosted), GitHub Actions (offline), TestCafe CLI | Set up offline Jenkins server on a local VM or Docker             |
| Clear Test Records               | Auditors need easy-to-read, time-stamped logs                 | ✅ Easy to use with default plugins      | Allure Reports, ExtentReports, Mocha + JSON exporters         | Add test reporters to export readable, timestamped reports        |
| Support Old + New Systems        | Tools must support COBOL + modern UIs like trading dashboards | ⚠ Needs hybrid skill teams              | Selenium (legacy UI), REST Assured, Cypress, SOAP UI          | Use separate scripts for legacy and REST-based systems            |
| Live Monitoring                  | Teams need real-time test visibility and alerts               | ✅ Easy with dashboard tools             | Grafana, Kibana, Jenkins Dashboards                           | Use ELK stack for log streaming, auto-update dashboards           |
| Fit Into Fast Development        | Must work inside CI/CD pipelines with minimal delays          | ✅ High – Plug into DevOps tools         | GitHub Actions, GitLab CI, Jenkins pipelines                  | Create pipelines that trigger tests on push/PR and notify results |
| Handle Secure Steps              | OTPs, 2FA, biometrics, payments must be testable              | ⚠ Needs test sandbox and mocks          | OTP Simulators, Stripe Sandbox, BrowserStack, biometric mocks | Use mock OTP/Stripe APIs and simulate inputs in test flows        |
| Fair Pricing                     | Tools must be affordable or offer flexible usage models       | ✅ Open-source tools are available       | Cypress, Playwright, Karate (all open-source)                 | Pick low-cost or free tools that offer essential functionality    |
| No Black Box AI                  | AI must explain what it does (for audits, trust, debugging)   | ⚠ Complex, needs explainability layer   | SHAP, LIME, LangChain, simple ML models                       | Wrap ML output in SHAP/LIME for clear justifications              |
| Data Privacy by Design           | QA must not expose sensitive user data during testing         | ⚠ Needs masking tools or synthetic data | Test Data Manager, Faker.js, data masking libraries           | Implement role-based access and synthetic data generators         |
| Resilience Testing               | Finance apps must tolerate failures gracefully                | ⚠ Needs isolated test environments      | Chaos Monkey, Gremlin, custom network simulators              | Run tests in staging to simulate network/service failures         |
| End-to-End Regulatory Simulation | QA must mimic full compliant workflows like KYC + AML         | ⚠ Requires deep workflow simulation     | Mock Regulatory APIs, sandbox KYC/AML tools                   | Simulate customer onboarding in isolated test flows               |



# Action Blueprint



|  QA Focus Area |  Real-World Risk if Ignored |  Who Must Act |  AI Automation Feasibility |  Feasibility in Real Banks (Low AI/Infra) |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Work Without Internet                                                                           | "Offline banking (e.g., ATMs) can't be tested in secure zones"                                               | IT Admin + QA Team                                                                               | ⚠ Low – Needs infra + local LLM setup                                                                         | ⚠ Possible with local infra; needs setup help                                                                                |
| Clear Test Records                                                                              | No traceability = audit/regulatory failure                                                                   | QA Lead                                                                                          | ✅ High – Auto-reporting & log generation                                                                      | ✅ Default plugins/logs work out-of-box                                                                                       |
| Old + New Systems Together                                                                      | "COBOL errors missed; UI fails go undetected"                                                                | Hybrid QA Team                                                                                   | ⚠ Medium – AI can assist in test case mapping                                                                 | ⚠ Needs mixed tech teams, partial AI use                                                                                     |
| Live Monitoring                                                                                 | No alerting = late bug fixes                                                                                 | DevOps + QA                                                                                      | ✅ High – AI can detect patterns, auto-alert anomalies                                                         | ✅ ELK/Grafana works even without AI                                                                                          |
| CI/CD Integration                                                                               | Releases break without test feedback                                                                         | QA + Developer                                                                                   | ✅ High – AI auto-runs regression tests                                                                        | ✅ Jenkins pipelines + AI optional                                                                                            |
| Secure Step Testing                                                                             | OTP/biometric not validated                                                                                  | QA Engineer                                                                                      | ⚠ Medium – Needs sandboxing & AI-driven mocks                                                                 | ⚠ Needs mock/sandbox, AI not mandatory                                                                                       |
| Fair Pricing                                                                                    | Tools too expensive for startups/small teams                                                                 | Budget Owner                                                                                     | ✅ High – Open-source AI/ML + free platforms                                                                   | ✅ Open-source tools work well                                                                                                |
| No Black Box AI                                                                                 | Opaque AI decisions can't be audited                                                                         | QA + ML Developer                                                                                | ⚠ Medium – Needs explainability (e.g., SHAP/LIME)                                                             | ⚠ AI explainability tools need ML awareness                                                                                  |
| Privacy by Design                                                                               | Data leakage during test sessions                                                                            | QA + Security Team                                                                               | ⚠ Medium – Use AI to generate synthetic test data                                                             | ⚠ Synthetic data gen needs dev effort                                                                                        |
| Resilience Testing                                                                              | Outages not handled gracefully                                                                               | QA + SRE                                                                                         | ⚠ Medium – AI can simulate failures, needs infra                                                              | ⚠ Needs infra to simulate failure                                                                                            |
| Regulatory Simulation                                                                           | QA misses legal KYC/AML flow coverage                                                                        | QA + Compliance Team                                                                             | ⚠ Low – Requires complex workflow emulation                                                                   | ⚠ Workflow emulation complex without APIs                                                                                    |

**Workflow Steps:**

- User initiates payment
- OTP is triggered
- AI test system captures OTP from simulator
- Inputs OTP into app
- Submits payment
- Auto-logs result with timestamp & screenshot
- Pushes result to dashboard + audit file

**Key Benefits:**

- No real payment involved (safe testing)
- Logs explain every step (audit-ready)
- AI alerts failure patterns (ML-enhanced testing)

| Task                     | AI Involvement                        |
|--------------------------|---------------------------------------|
| Simulate payment UI      | No AI – Basic scripted automation     |
| Generate OTP dynamically | Optional AI (generate/test coverage)  |
| Detect failure patterns  | ✔ AI – basic ML for test intelligence |
| Audit log & report       | No AI – default reporting             |
| Push to dashboard        | No AI – real-time logs only           |

# References & Source Links



## Startsida

Välkommen till Bankomat. Se våra valutakurser, nu flera valutor i ännu fler automater. Se på kartan var du hittar närmaste automat.



## Bankomat – Offline ATM QA

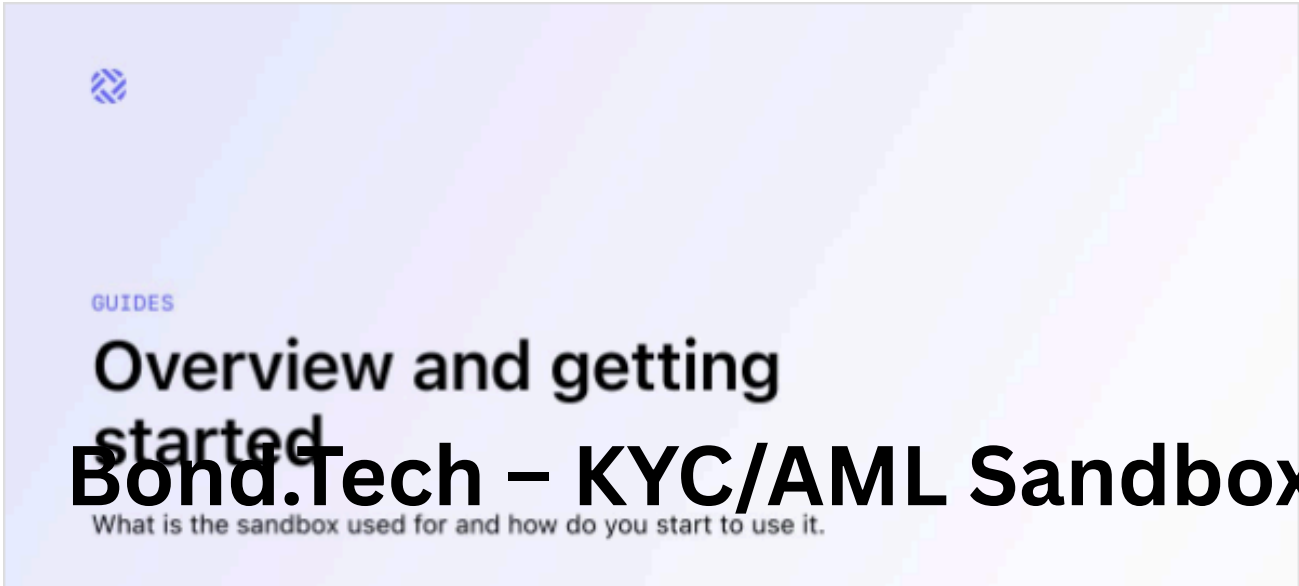


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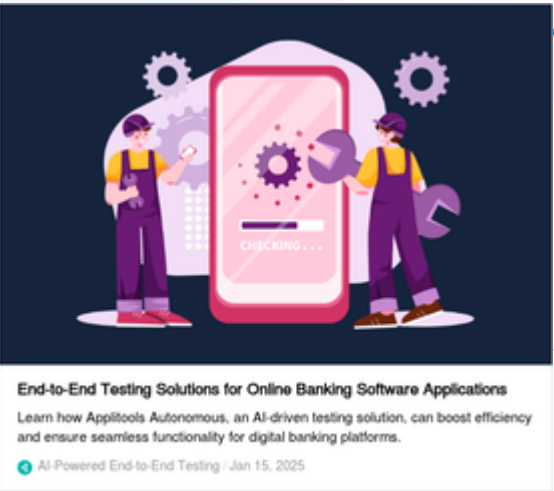
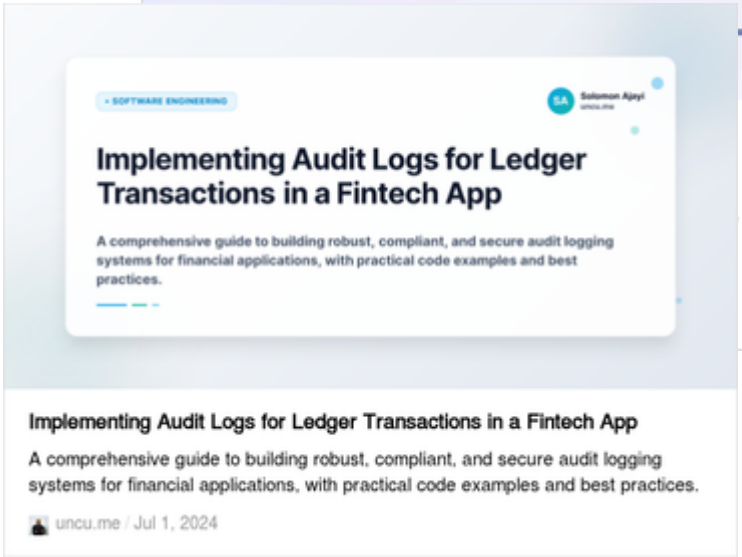
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## Fimatix – Real-time Test Dashboards



## Bond.Tech – KYC/AML Sandbox APIs





**Thank  
you**